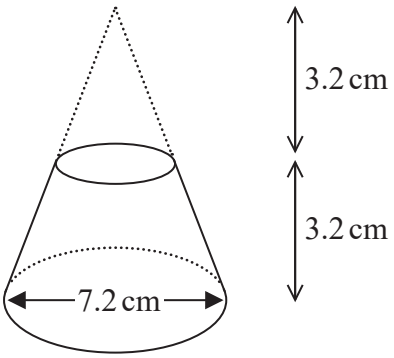
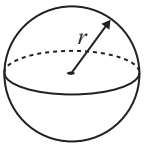
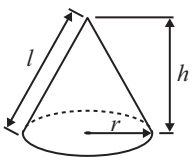
20 Here is a frustum of a cone.



Volume of sphere = $\frac{4}{3}\pi r^3$



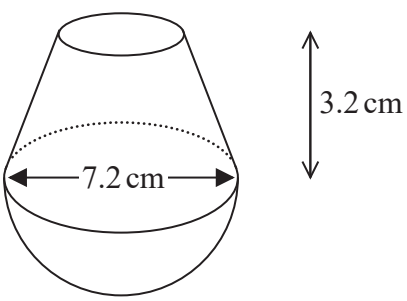
Volume of cone = $\frac{1}{3}\pi r^2 h$



The diagram shows that the frustum is made by removing a cone with height 3.2 cm from a solid cone with height 6.4 cm and base diameter 7.2 cm.

The frustum is joined to a solid hemisphere of diameter 7.2 cm to form the solid **S** shown below.

Solid **S**



The density of the frustum is 2.4 g/cm^3
The density of the hemisphere is 4.8 g/cm^3

Calculate the average density of solid **S**.

Large radius = 7.2 ÷ 2 = 3.6 cm.

The small and large cones are similar:

	Height	Radius	
	6.4	3.6	
÷2	3.2	1.8	÷2

1 mark
Subtract the cone volumes using the correct radii

Frustum volume = large cone – small cone

$\frac{1}{3}\pi \times 3.6^2 \times 6.4 - \frac{1}{3}\pi \times 1.8^2 \times 3.2$

$= (27.648 - 3.456)\pi = 24.192\pi$ cm^3

Hemisphere volume = $\frac{1}{2} \times \frac{4}{3} \pi \times 3.6^3 = 31.104\pi \text{ cm}^3$

1 mark
Use half the volume of a sphere

Frustum mass:

	cm^3	Grams	
	1	2.4	
$\times 24.192\pi$	24.192π	58.0608π	$\times 24.192\pi$

1 mark
Mass = volume × density

Hemisphere mass:

	cm^3	Grams	
	1	4.8	
$\times 31.104\pi$	31.104π	149.2992π	$\times 31.104\pi$

1 mark
Total mass ÷ total volume

Average density = $\frac{58.0608\pi + 149.2992\pi}{24.192\pi + 31.104\pi}$ = 3.75

1 mark
Average density 3.75 g/cm^3

3.75 g/cm^3

(Total for Question 20 is 5 marks)